

LOADZONE · PROBLEM STATEMENT

LoadZone problem statement

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	Reflect LDZ-DDR-001 (budget scope, installation, network) and the gateway distance found in LDZ-CAL-001
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (users, context, constraints, prior work, out of scope)
0.1	2026-09-25	Amish Chadha	Initial scaffold

LoadZone problem statement

Delivery drivers cannot see whether a loading bay is free before they reach it, so when the bay is taken they double-park in a traffic lane, a bike lane or a crosswalk, and cities have no routine record of how their loading bays are used. LoadZone aims to report the state of each vehicle slot in a loading bay, in real time and without cameras, at a cost low enough for a city, a business district or a community group to try on one street.

■ The problem in numbers

Loading bays in busy districts are full much of the day, and many of the vehicles in them are not delivering. In a 2017 observation study of five locations in downtown Seattle, passenger vehicles were 52 % of all vehicles parked in commercial vehicle load zones, and 41 % of commercial vehicles parked in unauthorized locations, rising to 55 to 65 % near retail centers ([Urban Freight Lab, 2019](#)). The same research group reports that commercial vehicle load zones often reach 90 % occupancy or more during business hours ([Urban Freight Lab, Final 50 Feet](#)).

Stops are short, so information has to be quick. In the Seattle study, 60 % of delivery vehicles parked for 15 minutes or less and 81 % for 30 minutes or less ([Urban Freight Lab, 2019](#)). A bay can change state several times an hour, which static signs and occasional manual surveys cannot capture.

Demand keeps growing. Online sales rose from 16 % to 19 % of all retail sales in 2020 ([UNCTAD, 2021](#)), and in Great Britain van traffic rose 2.3 % to 59.0 billion vehicle miles in the year to September 2024 ([Department for Transport](#)).

■ Users and context

User	Need
Delivery and service drivers	Know, before arriving, which bay or slot is free; avoid circling and double parking
City curb and transport teams	Occupancy, turnover and dwell time by bay and hour, to set bay size, hours and pricing
Parking enforcement	Know which bays have overstays without a patrol or a camera
Shops and business districts	Shared loading space that works for their suppliers; evidence to request more bays
Cyclists and pedestrians	Fewer vans stopped in bike lanes and at crossings

The setting is an on-street loading bay along a curb, typically long enough for one to three vans, on a street with shops, offices or homes. The street may already have poles (lighting or signs) that a city allows sensors to be fixed to. A LoRaWAN gateway may already exist nearby (a city network, The Things Network or the lab's TwinKit gateway), or one must be added. From road level under a parked van a puck reaches only about 350 m in a street canyon (LDZ-CAL-001), so a gateway within 300 m of the bay is required (R4 as restated, LDZ-DDR-002).

■ Prior work

- **Parking occupancy sensing is established.** San Francisco's SFpark pilot (2011 to 2013) collected and published real-time availability for 7,000 of the city's 28,800 metered spaces ([SFMTA](#)). Commercial in-ground and surface parking sensors are sold today, usually as closed devices tied to a vendor's platform and subscription.

- **Curb data standards exist.** The Open Mobility Foundation's Curb Data Specification defines a Curbs API for bay locations and rules, an Events API that can take events from sensors, and a Metrics API for dwell time and occupancy ([Open Mobility Foundation](#)). An open sensor that can feed this format lets cities compare sensor data with carrier data.
- **Cities are testing digital curbs.** In 2024 a collaborative of cities funded through the US Department of Transportation SMART grants program began comparative research on curb infrastructure, policy and demand, with the Urban Freight Lab ([Urban Freight Lab](#)).
- **Low-power magnetometers are cheap and common.** Parts such as the LIS2MDL are 3-axis, 16-bit sensors with a ± 50 gauss range, a single-measurement mode and an operating range of -40 to +85 °C ([STMicroelectronics](#)).
- **The lab's FieldNode** provides the LoRaWAN radio core, firmware base and, for the sign option, the solar power and pole mount.

What is missing is an open, inspectable bay sensor that a small city or a community group can build, bond to the road and connect to any LoRaWAN server, with data in an open curb format and no camera.

■ Constraints

- Garage-buildable prototype, \$120 USD or less for a two-slot bay (two pucks). The optional sign and its host FieldNode are costed separately and are outside this budget (LDZ-DDR-001, D6, decided by Amish on 2026-09-25).
- No cameras or microphones; the device may report only slot state (free or occupied), timestamps and device health.
- No cutting or coring of the road for the prototype; the puck is bonded to the surface with two-part road-marker epoxy and is removable.
- Installation only with the road authority's permit, by a trained crew under traffic management.
- Battery powered for at least five years; no wiring in the road.
- Uses the FieldNode radio core so firmware and tools are shared across the lab.
- The sign option requires a private network server (such as TwinKit) or a city network that allows about 200 downlinks a day (decided, LDZ-DDR-002); The Things Network allows 10 per device ([TTN](#)).
- Must survive being driven over by loaded delivery trucks, road salt, water, oil and summer road temperatures.

■ Out of scope

- Payment, booking or reservation of bays (the data can feed such systems later).
- Enforcement actions or identifying vehicles; LoadZone never reads number plates.
- Passenger car parking guidance at scale (the same puck could do it, but it is not the target).
- In-ground (cored) installation, which needs road works; noted as a later variant.

■ User research and co-design

This design is for communities the author is not part of, so requirements come from the people who will use it.

- [] Identify a local partner organization (Helpful Engineering network, NGO or university)
- [] Run co-design sessions with intended users; record who, where and what was learned
- [] Validate load, distance, terrain and cost assumptions in the field

- [] Revise requirements (REQ) from findings before freezing the design

For LoadZone the intended users are delivery drivers, a city curb or transport team and the shops on one test street. Questions for them: would drivers look at a sign, a map layer or neither; which bay lengths and slot splits are typical; and which data the city needs to act on.